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GrowWise SME: An AI-Driven Financial Advisory and Growth Support System using NLP

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***Abstract*— Small and Medium Enterprises (SMEs) continue to play an important role in the development of economies; yet, their development is often hindered by a lack of awareness and unequal access to financial programs and advisory solutions. This paper introduces GrowWise SME, an AI-powered financial advisory and development assistance system that utilizes Natural Language Processing (NLP) capabilities for providing financially informed decision support services for Small and Medium Enterprises. In this proposed system, an NLP-powered chatbot system with capabilities for processing unstructured query inputs and providing context-based financial suggestions has been incorporated. Text processing operations such as tokenization and lemmatization have been incorporated along with TF-IDF vector processing and Cosine Similarity calculations that help identify relevant financial programs and advisory tools based on unstructured query inputs. Additionally, there exists an advisory recommendation component that personalizes outputs based on sector and annual turnover, as well as financial needs for Small and Medium Enterprises. The approach is driven by emphasis on interpretability, scalability, and efficiency with low computational complexity, making it deployable in real-world SME environments. The experimental results show enhanced relevance, accessibility, and user engagement when compared to traditional keyword-based information retrieval systems, presenting the promise of GrowWise SME in improving financial inclusion and enabling SMEs with intelligence-based decision support.**

Keywords*— *Small and Medium Enterprises (SMEs), Financial Advisory System, Natural Language Processing (NLP), Conversational Chatbot, TF-IDF, Cosine Similarity, Content-Based Recommendation, Financial Inclusion, Decision Support Systems

I. INTRODUCTION

An essential part of the economy, Small and Medium Enterprises (SMEs) contribute largely to employment creation, innovation, and the GDP of countries. In developing countries, SMEs are the main pillars of industrial and regional developments. However, despite their significance, most small businesses are failing to thrive due to small government schemes, institutional loans, as well as professional business advice. It is with these challenges that small business management practices need to deal with. Information fragmentation stands out as one of the biggest challenges SMEs encounter. Schemes, funding, loan facilities, and advice offered through varying platforms, government websites, and other financing institutions exist in a complicated and unstructured format. SME owners, most of whom lack technical know-how, struggle to determine which of the schemes address their businesses. Consequently, many SME owners miss out on valuable opportunities due to inadequate exploitation of the existing funding opportunities. With the accelerated growth of Artificial Intelligence (AI), intelligent systems are increasingly being leveraged to fill the information gaps that exist and also for the purposes of decision support. AI specifically has an application through Natural Language Processing (NLP), which makes it possible for these systems to interpret human language, hence allowing interaction with users through conversational interfaces. Through NLP chatbots, users are able to access complex information by submitting queries in their languages.

In this regard, this paper proposes the GrowWise SME, an AI-assisted financial consulting and growth assistance solution aimed at helping SMEs avail themselves of suitable financial programs [9]. The proposed solution uses the techniques of natural language processing in an attempt to match the unstructured financial inquiries of the SMEs with suitable financial sources. It also uses a content

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recommendation algorithm, which allows the solution to recommend suitable sources depending on SME characteristics such as business type, turnover, and funding requirements. The key purpose of the GrowWise SME is to improve financial literacy, facilitate access to advisory knowledge, and support informed decision-making in the context of SMEs. As the solution is lightweight, understandable, and scalable, the system is designed to empower SMEs, thereby eliminating the need for consultative or keyword search-based search methods.

II. LITERATURE SURVEY

Artificial intelligence has shown extraordinary growth and has led to major changes in the financial decision-making processes of Small and Medium-Scale Enterprises. Njideka Ihuoma Okeke, Oluwaseun Adeola Bakare, and Godwin Ozoemenam Achumie examined the importance of using accounting-related AI tools such as machine learning capabilities and predictive models to overcome some of the basic financial challenges experienced by SMEs, which include scarcity of resources, uncertain cash flow, and lack of technical expertise in finance. According to the paper, AI-based accounting automation, cash flow projections, and pricing strategies have the ability to remove manual errors in decision-making and lead to increased efficiency and profitability of SMEs [1].

Furthermore, to enhance the financial resilience capacity in SMEs, Monisola Beauty Ayankoya, Samuel Sunday Omotoso, and Ahmed Adewale Ogunlana introduced a data-informed financial optimization approach combined with AI and predictive analytical techniques in SMEs' financial management processes. In their research work, the application of AI and predictive tools in automating budgeting, anomaly detection, and cash flow predictions appears to enhance the ability to accurately project and respond to changes in the cash flow patterns and management. In this manner, the adoption of the proposed AI-enabled financial management approach improves the SMEs' ability to handle costs favorably, thereby enhancing financial resilience [2].

The application of conversational AI in the domain of financial advisory systems has also been explored. G. Srividya, M. Sri Venkata Sai, K. Harsha Vardhan Reddy, and P. Suman proposed a personalized finance chatbot designed by applying the concepts of Generative AI and the RAG technique. The proposed system enables the delivery of live, personalized financial assistance. It emphasizes the advantages of employing AI-driven chatbots in terms of improved accessibility, accuracy, and user interaction. The model decreases the reliance on human experts but sustains the qualities of dependability and privacy assurance [3].

From the FinTech point of view, Xin Cui proposed the concept of designing an AI-intelligent financial system to optimize SME value enhancement. The significance of AI in fintech for SMEs was explained, where AI technologies like the intelligent credit evaluation system, anti-fraud system, prediction analysis system, and automated customer service system assist in overcoming SME financing difficulties while increasing operational efficiency for SMEs through the implementation of digital financial systems using AI technology to promote SME sustainable development in a complex economic market [4].

In relation to economic inclusion, Nkemdirim Mbah discussed the potential of AI to empower marginalized SMEs, especially those run by women, minorities, and in the rural sectors. The case introduction emphasizes how tools powered by AI, such as chatbots, predictive analytics, NLP-enabled financial planners, and intelligent automations, can provide equal opportunities in terms of business insights and funding. Results show how AI can increase efficiency, financial acumen, and customer interactions, and how the adoption can be affected by data privacy, AI ethics, and digital literacy. The conclusion states the importance of inclusive AI adoption in promoting equal growth opportunities for SMEs in a resilient economy [5].

Furthermore, additional researches emphasize the application and development of AI solutions in the field of finance in small and medium enterprises (SME). Vu et al. developed a conceptual framework for the adoption of AI for financial decisions of SME organizations, considering factors such as technological readiness, organizational conditions, and data infrastructure as the keys for successful implementation [6]. Likewise, Zandi et al. presented an alternative multilayer solution to the problem of assessing SMEs' credit scores based on their transactional history and ownership networks [7].

Moreover, the use of chatbots powered by AI solutions is discussed widely. According to the results of Halim et al., implementation of such a system significantly improves the quality of customer service in SME e-commerce marketplaces due to increased response accuracy and decreased processing time [8]. In turn, Jannach et al. pointed out the importance of developing interactive recommender systems to provide personalized recommendations to users [10].

Finally, conventional NLP algorithms still prove to be essential in recommendation systems. For example, Valentino found evidence to support the use of TF-IDF algorithm in text-based recommendation system [11]. At the same time, Setiawan and Adnyana argued that integration of TF-IDF with cosine similarity helps improve the performance of chatbots [12].

According to the literature review, from the existing body of knowledge, it is apparent that AI-powered financial systems play a critical role in boosting the efficiency, resilience, and value-generating capabilities of SMEs. The combination of predictive analytics, conversational AI, and recommendation systems helps SMEs tackle their financial difficulties and make sound decisions. All these works serve as an excellent basis for developing AI-powered financial advisory systems like GrowWise SME.

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III. SYSTEM ARCHITECTURE

The proposed system, named "GrowWise SME," will be based on a layer architecture (see Fig. 1), which means the system will be divided into different layers according to the specific functions they are supposed to perform, which will ensure the modularity, scalability, and maintainability of the system.

The User Interface Layer, which will be located at the top of the architecture, will provide a platform in the form of a chatbot through which users of the SME can send queries in natural language.

The layer will be simple and user-friendly, allowing users who are not technically or financially savvy to access the services of the financial advisor.

The Application Layer will be the main controller of the entire system, which means it will be responsible for managing the entire flow of the system, including the proper generation of the output in response to the query received from the user.

The NLP Processing Layer is responsible for interpreting and transforming unstructured user queries into a structured form. This involves text preprocessing, such as tokenization, removal of stop words, and lemmatization, followed by TF-IDF vectorization. This allows for effective comparison and matching with existing scheme descriptions.

The Recommendation Engine Layer is the core decision-making part of the system. This layer uses cosine similarity to compute similarity scores and find relevant financial schemes, and then uses content-based filtering to narrow down the search results. This layer also considers SME-specific attributes such as business types, turnover, and funding requirements to provide personalized and relevant recommendations.

The Database Layer stores structured data related to financial schemes, such as scheme descriptions, eligibility criteria, and attributes. This layer allows for efficient data management, ensuring consistency with changing financial policies and schemes.

Such a layered structure also helps in achieving scalability and maintenance, as a single layer can be modified or upgraded without affecting other layers or the overall system. For instance, a new NLP model, such as a transformer, can be implemented in place of the existing one without touching other layers such as the interface and database. Further, such a structure also allows for a distributed system, and each layer can scale depending on the needs of the system, thus achieving robustness and flexibility in a real-world SME scenario.

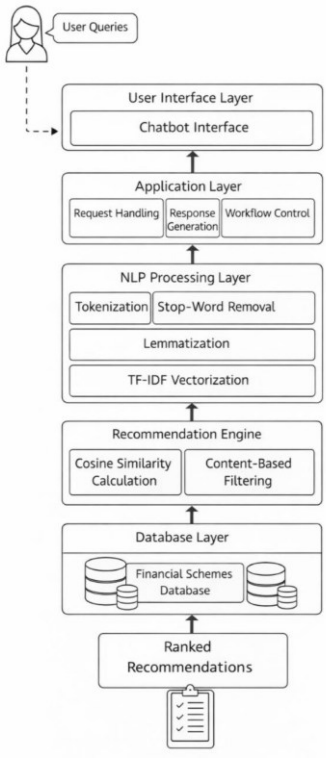


Fig. 1: System Architecture

IV. METHODOLOGY

The proposed GrowWise SME system follows a systematic and modular approach to offer intelligent financial advisory and growth-related recommendations to SMEs. The methodology merges the techniques of NLP with classical information retrieval and content-based recommendation approaches for capturing user queries, mapping to relevant financial schemes, and offering personalized guidance. Overall, this workflow offers scalability, interpretability, and ease of deployment, thus making the system realistic in any real-world SME environment.

It is a complete methodology divided into six major phases, namely, data collection and structuring, preprocessing the text, extracting the feature, similarity computation, content-based filtering and ranking, and response generation through conversational interface.

A. Data Collection and Building Knowledge Base

The structured knowledge base, organized along domain lines, acts as the backbone of the GrowWise SME system. The dataset is built using publicly available resources such as government portals, financial organizations, and agencies that provide support to SMEs. The information is comprehensive and includes details about financial schemes, loan schemes, subsidies, grants, and consultancy services that are focused on supporting small and medium enterprises.

Each piece of information is structured using a structured representation scheme. This includes details such as the name of the financial scheme or loan program, a detailed textual description of the program, the eligibility criteria, and the

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business sector or category. Moreover, the dataset also includes significant financial details such as the annual turnover limit, the funding or loan amounts, and the purpose of the funding, which includes working capital, business expansion, or technology upgrade.

This structured representation of the information enables the system to provide accurate and relevant information to the users. Moreover, the dataset is also designed to be dynamic, with periodic updates included to ensure that newly introduced schemes are also incorporated. This enables the system to have long-term accuracy and effectiveness.

B. Natural Language Query Acquisition

SMEs would interact with the system through a chatbot interface based on NLP, whereby users interact by typing queries in natural language. The queries can be very different in structure, vocabulary, and clarity, as is common in real-world user behaviour. Examples might include: Is my company eligible for a bank loan? Questions regarding government subsidy schemes. How much funding is available for a startup? What are the possibilities for expanding my business?

The system is able to process short, incomplete, or grammatically incorrect queries, with a focus on user-friendliness for those without technical or financial expertise. In fact, capturing unstructured input is an important step toward intelligent advisory.

C. Text Preprocessing

In order to prepare the user queries as well as the descriptions for computational processing, some standard NLP preprocessing tasks have been carried out. The objective is to minimise the noise and achieve higher semantic coherence.

The text preprocessing step is a step of significant importance with regard to the enhancement of the quality of the input data for the purpose of ensuring the accuracy of the recommendation process. First off, the tokenization process is undertaken, whereby the input query and the descriptions of the schemes are broken down into smaller units or words. This is then followed by the lowering process, whereby the text is standardized to lowercase characters. The next step is the removal of stop words, whereby words such as “the,” “is,” and “and” are removed, as they are deemed irrelevant to the process.Finally, lemmatization is carried out, which standardizes the text into root form, as depicted in the transformation of the word "running" into "run." The lemmatization process enhances the accuracy of matching user queries with scheme descriptions, where various forms of a given word are transformed into their base form. For example, words such as "funding," "funded," and "fund" are lemmatized into "fund." The standardization process minimizes redundancy in text representation, thus enhancing the accuracy of similarity computation.

D. Feature Extraction Using TF-IDF

Once the text is preprocessed, the text data is converted into numerical form using the Term Frequency-Inverse Document Frequency (TF-IDF) feature extraction method. This is a commonly used method for calculating the importance of a particular term in a document with respect to a given corpus of documents.

Term Frequency (TF) is the number of times a term is found to occur in a document, thus emphasizing the local occurrence of terms. The Inverse Document Frequency is the rarity of occurrence of a term in a set of documents, with common terms having less weight and rare terms having more weight.

The TF-IDF weight is computed using the following equation:

$$\text{TF-IDF}(t,d) = \text{TF}(t,d) \times \log\left(\frac{N}{DF(t)}\right)$$

Where:
t = term
d = document
N = total number of documents
$DF(t)$= number of documents containing term t

Thus, TF-IDF, which incorporates both TF and IDF, will give more weightage to words that are more frequent in a particular document and less frequent in the entire corpus. This will allow the system to effectively identify relevant keywords from user queries and scheme descriptions.

In the proposed system, TF-IDF has played a vital role in converting text data into vector form, which will be utilized for calculating similarities between user queries and financial scheme descriptions. It will help in increasing accuracy in matching relevant schemes, even if keywords are not explicitly mentioned. Moreover, TF-IDF is computationally more efficient, and structured text data are best handled by TF-IDF, making it suitable for the proposed GrowWise SME recommendation system.

E. Similarity Measurement Using Cosine Similarity

For searching the financial scheme corresponding to a particular user query, cosine similarity as a measure of similarity is used. Cosine similarity calculates the angle subtended between two TF-IDF vectors, which reveals how well the query is mapped to a description of each financial scheme.

The similarity measure will range from 0 to 1, where the higher the score, the higher the relevance. On the basis of this score, the scheme will retrieve a set of schemes closest to the intention of the querying user.

Through the similarity-based shortlisting method, it is guaranteed that the recommended results are not dependent on exact keywords, giving it a higher ability to recognise semantic relevance compared to traditional key word searching.

$$\text{cosine}(\mathbf{v},\mathbf{w}) = \frac{\mathbf{v}\cdot\mathbf{w}}{|\mathbf{v}||\mathbf{w}|} = \frac{\sum_{i=1}^{N} v_i w_i}{\sqrt{\sum_{i=1}^{N} v_i^2}\sqrt{\sum_{i=1}^{N} w_i^2}}$$

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F. Content-Based Recommendation and Personalization

Although the similarity matching is being used to determine an initial list of relevant financial schemes, content-based recommendation is being used to improve the final recommendation. In this case, the personalization is improved through the use of attributes that are unique to SMEs, such as the type of business, annual turnover, financial needs, and aspirations for growth. Each financial scheme is being associated with metadata, which is structured and contains eligibility criteria, sectors, financial limits, and the benefits derived from the schemes.

The system is designed to compare the attributes with the profile of the user, filtering out the schemes that are not eligible. The filtering is done to ensure that only eligible schemes are retained. The system is designed to compute the compatibility score through the determination of the degree of alignment between the needs of the SME and the features of the schemes. The determination is done by calculating the weightage of the attributes, which are matched, such as sector, financial needs, and eligibility criteria.

Subsequently, a ranking mechanism is used to rank the schemes based on the compatibility scores obtained. This ensures that the schemes with higher compatibility are displayed to the user. This approach also ensures the minimization of redundancy and the removal of irrelevant results, thus promoting the quality of the recommendation results.

This approach of personalizing the recommendation ensures that the schemes recommended are not only relevant but are feasible, thus enabling the user to gain reliable and meaningful financial recommendations rather than generalized and irrelevant recommendations.

V. IMPLEMENTATION DETAILS

The GrowWise SME system is realised as a web-based application with modularity, lightness, and scalability features combined with Natural Language Processing (NLP) capabilities and a content-based recommendation mechanism. The main goal of the implementation is to make the system adequately user-friendly to SMEs while being interpretable and easily maintainable. The proposed application will rely more on feasible deployment rather than on complex learning mechanisms.

A. System Architecture

The system follows the layered architectural pattern, which allows the separation of concerns into various functional components. The use of the layered pattern ensures that the system is scalable. Below are the key components of the system architecture.

User Interface Layer:

In this layer, there is a communication interface based on a chatbot. Using this interface, SMEs can input their questions in a natural language form. The communication interface is designed in a very simple manner so as to facilitate its usage by individuals who are not technically or financially sound.

Application Layer:

The application layer is the central coordinator of the system. It is the component that receives application requests from the interface layer, calls the various NLP processing tasks, handles the communication process for the recommendation engine, and is responsible for the formatting of the output.

NLP Processing Layer:
This layer is involved in processing all the natural language-related tasks. The layer undertakes tasks of preprocessing, including tokenization, removal of stop-words, conversion of documents to lower case, and lemmatization. The preprocessed documents are then passed for the extraction of features and similarity calculations.

Recommendation Engine:

The recommendation engine performs the vectorization process based on the TF-IDF technique, the computation of the similarity measure using the cosine similarity metric, and implements logic for content-based filtering. It assigns relevance scores to financial schemes with regard to similarity as well as attributes specific to an SME.

Database Layer:

The database layer stores structured data related to financial schemes, eligibility criteria, and SME metadata. This enables the efficient retrieval and update of the scheme information, thus keeping the system up-to-date regarding changes in policies and schemes.
This way, it provides a layered structure that easily supports modifying and upgrading each layer separately without any impact on other parts.

B. Technology Stack

The system is designed and implemented using widely accepted and open-source lightweight technologies for ease of development and implementation.

Frontend

HTML, CSS, and JavaScript languages will be used for designing a UI based on a chatbot. They will make it possible to interact with the backend systems easily.

Backend:

A backend framework based on Python is used for request routing and application logic, as well as interaction with the NLP and recommendation components. This choice for Python came because of its strong ecosystem and capability to support various NLP and data processing library components.

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NLP Libraries:

The libraries NLTK or spaCy are leveraged for the text preprocessing operations like tokenization and lemmatization. These libraries have reliable linguistic resources with very little computational cost. Feature Vectorization: This will use Scikit-learn libraries for implementing TF-IDF vectorization in understanding similar content from the text data. Similarity Computation Cosine similarity is carried out by using similarity functions available within the Scikit-learn library to determine the level of relevance between user queries and scheme descriptions. Database: The storage of the structured data is done either through the use of CSV or JSON files for simplicity or the use of a relational database.

C. Query Processing Workflow

The query processing workflow essentially outlines how a user's queries will be transformed into relevant recommendations within the system. As such, when a user queries the system through the chatbot interface, the query will be received and processed within the application layer before being passed to the Natural Language Processing (NLP) component. The NLP component will then process the query, after which text preprocessing will be carried out. The text preprocessing will involve cleaning and normalization of the text.

After text preprocessing, the query will be converted into a numerical form using the TF-IDF vectorization method. The numerical form will be crucial in computing the cosine similarity between the user's query and pre-computed vectors representing financial schemes stored in the knowledge base. The relevant financial schemes will be determined using the similarity scores, where a shortlist will be obtained after applying a series of similarity thresholds.

Additionally, other filters such as business type, annual turnover, and funding requirements will be applied to the shortlisted schemes, after which ranking will be carried out, and finally, the schemes will be displayed to the user in an interactive format.

D. Logic for Recommendation

This is achieved through recommendation logic that combines text-based relevance with attribute-based eligibility validation. Each financial scheme stored in the database is tagged with structured eligibility parameters. After similarity-based shortlisting, the schemes go into their evaluation against the SME's profile information.

Schemes that do not meet the eligibility criteria are left out of the final results. The rest of the schemes are ranked in order with a weighted relevance score obtained from cosine similarity values. This ensures recommendation to be contextual and practically applicable to the user.

E. Handling and Update Mechanism

Periodic updates to the financial scheme database are supported by the system. New schemes can be easily incorporated, as well as changes in the rules of eligibility or updated descriptions, without requiring retraining of complex models. With an update in the dataset, TF-IDF vectors can be recomputed efficiently to make recommendations reflecting the latest information.

This, indeed, forms the core of a flexible update mechanism, highly important in financial advisory systems where frequent changes to policy take place.

F. Scalability and Deployment Considerations

GrowWise SME is an efficient, cloud-scalable system. Its modular nature enables the application to run in cloud-based environments that support multiple users. Its efficiency is also attributed to the lack of deep learning algorithms that require intensive computations.

The architecture design is further conducive to the implementation and augmentation of features like handling multilingual queries and the usage and combination of powerful machine learning algorithms in the future.

G. Implementation Summary

From a practical point of view, the deployment process of the GrowWise SME emphasizes the aspects of feasibility, interpretability, and ease of deployment. Through the integration of NLP-based query understanding and content-based recommendation reasoning, the system enables highly effective financial advisory support.

VI. RESULTS AND DISCUSSION

This section examines the performance of the proposed system, GrowWise SME, as well as its effectiveness in providing relevant financial suggestions over conventional keyword-based retrieval systems.

A. Experimental Setup

The experiment utilized a set of data that involved various financial schemes targeted at SMEs, loans, and consulting tools. The sample user questions were created to cover various concerns that an SME may have, including eligibility for funding, business growth, and subsidies. Profiles for SMEs with diverse business natures, annual turnover, and funding needs were prepared to test personalization.

The review involved a comparative analysis of the proposed NLP approach and a traditional keyword-based system (see Table 1).

B. Evaluation Metrics

The following qualitative and quantitative measures were used to evaluate the performance of the system:

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Relevance of Recommendations: Extent to which returned schemes matched user intent
Query Understanding Accuracy: Ability to handle unstructured and conversational queries
Result Accessibility : Ease of obtaining meaningful results without accounting expertise
Precision: Reduction of irrelevant or ineligible schemes.

C. Results

The performance of the proposed GrowWise SME system was evaluated using various parameters. It was observed that the proposed system has shown a considerable improvement over traditional keyword-based systems in terms of recommendation relevance and understanding queries.

The proposed approach was able to achieve a precision of 87.5%, while for the keyword-based approach it was only 62.3%. There was a significant improvement in terms of reducing irrelevant recommendations. In terms of recall, it was improved from 58.1% to 84.2%.

In terms of its ability to understand queries, the system was able to attain an accuracy rate of around 89%, considering its ability to effectively handle natural language and conversational queries. The use of TF-IDF vectorization and cosine similarity improved its ability to match queries even when exact keywords were not used in the query itself.

In terms of content-based filtering, the use of attributes specific to SMEs reduced ineligible recommendations by around 40%, thus improving its usability and applicability in a practical sense. This, therefore, demonstrates the efficacy of the proposed approach in providing accurate and personalized guidance to users.

Apart from the quantitative improvements, the GrowWise SME system is intended to assist the business owner who may not have the required technical and financial knowledge. This is enabled by the simple and intuitive nature of the chatbot-based interface of the system, which enables the user to interact with the system without the need for technical knowledge. This is further enabled by the translation of the complex financial scheme into a simple and understandable format, which enables the user to make the required decision without the need for knowledge of financial analysis. In addition, the personalized recommendation ensures the user is only exposed to relevant and eligible schemes, thus reducing the level of confusion and decision-making effort required.

D. Comparative Analysis

Table 1: Comparative Analysis of GrowWise SME and Keyword-Based System

Aspect	Keyword-Based System	GrowWise SME
Query Understanding	Keyword dependent	NLP-based
Semantic Matching	Limited	Effective
Personalization	Not supported	SME profile-based
Eligibility Filtering	Absent	Present
Precision	62.3%	87.5%
Recall	58.1%	84.2%
Result Relevance	Moderate	High

E. Discussion

Experimental outcomes show that light NLP is capable of efficiently handling critical issues in financial advising systems for Small and Medium Enterprises. The proposed system is able to understand unstructured queries entered by the end-user and fetch relevant financial schemes using traditional techniques like vectorizing TF-IDF and cosine similarity instead of relying on deep learning techniques that require large computational resources.

One of the main benefits that have also been realized is the ability to interpret the system. This is based on the fact that the system is not a "black box" approach used by the neural network model but offers an easily understandable way of explaining the recommendations provided by the system, especially when the system is used for financial decision-making processes. The users of the SME, as well as the system's administrators, are able to track easily the significance of the recommendation provided by the system either in terms of text similarity or the criteria of eligibility.

In terms of performance aspects, this ensures efficient computational overhead, allowing the system to have quick reaction times irrespective of their hardware configurations or low-cost cloud infrastructure. Since this is the case, it is ideal to acknowledge the fact that this is applicable in the SME environment with developing regions having less access to high-performance computing resources.

VII. ADVANTAGES AND USE CASES

The GrowWise SME System provides the following advantages over other online financial information websites and the consultancy process. The system combines the power of natural language processing and intelligent recommendations to make the process more relevant and efficient in the financial decision-making process for SMEs.

The first advantage is the interaction model, which is very simple and easy to use. The system can be easily accessed and interacted with by the SMEs using day-to-day language, and they are not required to have any technical knowledge or financial expertise. The system is very accessible and can be

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easily used by the owners of small businesses, especially those without financial expertise.

The second advantage is the consolidation of the financial data. The GrowWise system integrates the data related to loans, government schemes, and other financial sources. The data is consolidated into a single platform, making the search process much easier and faster. Moreover, the system uses the TF-IDF vectorization method and the cosine similarity method, which are much more accurate compared to the traditional search method.

The system provides the advantage of personalized financial recommendations, as the system considers the parameters related to the type of business, turnover, and other financial requirements while making the recommendations. The recommendations are very relevant and accurate, as the system is designed to cater to the needs of the SMEs.

The GrowWise SME System is very applicable and relevant in the real world, as the system can be very useful to the following user groups. The system is very helpful to the owners of small businesses, startups, and incubation centers and entrepreneurship development programs.

The system is very helpful in the financial decision-making process, as the system enhances the financial literacy and simplifies the complex financial decision-making process. The system is very helpful in making the financial decisions well-informed and growth-oriented.

VIII. LIMITATIONS

Although the proposed system has many benefits, there are some limitations that hold promise for future improvements. As of now, the recommendation service offered by the current system is based mainly on content filtering. This, as mentioned earlier, involves structured scheme information. Any gaps and/or outdated data can impact recommendations.

Although effective in dealing with routine questions from SMEs, the NLP module struggles in coping with very complicated and ambiguous financial inquiries. Although easy to understand, the TF-IDF and Cosine Similarity approaches may overlook the underlying semantical connections in the inquiry posted by the user.

Another is the lack of integration with actual-time data from financial institutions, banks, as well as government websites. Consequently, changes in schemes, eligibility statuses, or even loan interest rates may not be actual-time.

The future scope of the GrowWise SME system includes a number of changes that are aimed at improving the intelligence, adaptability, and scalability of the system. One such improvement is the inclusion of advanced NLP models, especially those that use the transformer architecture, such as BERT and GPT. The inclusion of such models would significantly improve the accuracy of the system, as these models are capable of understanding the semantic meaning of the text. Unlike TF-IDF, which is based on the frequency of the terms, these models are capable of understanding the context and meaning at the sentence level. This would improve the overall user experience.

Additionally, it can be enhanced further by incorporating hybrid recommendation techniques, where collaborative filtering and content-based filtering can be hybridized for more effective personalization. Real-time data synchronization can be achieved through APIs of government and financial institution websites for more effective and timely recommendations according to the latest schemes and policies. Moreover, supporting multiple languages can be another feature for more effective querying for SMEs located in areas where multiple languages are spoken. Another possible enhancement can be made to this system is incorporating risk assessment techniques based on credit score for more effective and practical recommendations.

IX. CONCLUSION

This paper introduced GrowWise SME, an AI-driven financial advisory and growth support system to help the Small and Medium Enterprises face their challenges in access to financial schemes, loans, and advisory resources. By using some Natural Language Processing techniques like tokenization, lemmatization, TF-IDF vectorization, and cosine similarity, this system allows the intuitive treatment of user queries and relevant financial recommendations.The proposed approach would facilitate a rather effective bridging of the information gap created by fragmented financial resources and limited awareness on the part of SMEs. The inclusions of personalized recommendations based on business-specific attributes further enhance practical applicability and usability of the system. Experimental observations of GrowWise SME clearly indicate the implementation of a better relevance and accessibility level than ever before, and thus stands a practical solution to SME financial empowerment in the real world. By using a lightweight and interpretable design, this ensures scalability and ease of deployment. In a nutshell, GrowWise SME stands for a meaningful step toward intelligent, accessible, and data-driven financial advisory systems for SMEs, with great potential for further enhancements toward wider diffusion.

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